

Experiment 3: Buckling of columns

Objectives

Understanding the phenomenon of column buckling, and measuring experimentally the buckling (or critical) load for columns under different boundary conditions.

Background

Slender structures often fail in one of two ways: 1.) Stresses reaching a value where the material fails, 2.) A sudden violent change in the shape of the structure caused by a loss in stability of its equilibrium configuration. In this experiment we will study the latter mode of failure, referred to as buckling.

To understand the phenomenon of buckling adequately, you must answer the following questions:

- 1.) What is the precise definition of buckling?
- 2.) Do the terms “buckling” and “bending” refer to the same phenomenon? If not, what is the difference between the two?
- 3.) Can a slender body buckle without bending? Can it bend without buckling?
- 4.) The critical axial load P_{cr} at which an inextensible slender column of length L , Young's Modulus E , and second moment of area I buckles is given by,

$$P_{cr} = \frac{\pi^2 EI}{(KL)^2}$$

Where, K is an effective length factor that depends on the boundary conditions on the column as:

1. Both ends pinned, $K = 1.0$
2. Both ends fixed, $K = 0.5$
3. One end fixed, and the other pinned, $K \approx 0.699$
4. One end fixed, and the other end free, $K = 2.0$

Which of the parameters in the formula above are material in nature, and which ones are geometric?

5.) Assume a column of a circular cross-section of radius R and length L . If we wish to increase length of the column to $2L$, while keeping the critical load P_{cr} the same, to what value should we change R ?

Some online resources

You may use the following videos to refresh your understanding of buckling in columns

[Understanding Buckling](#) (14:48 minutes)

The experiment

Task 1:

You will be provided with an apparatus in the ME lab where you can load a slender column incrementally under different boundary conditions. You must load the column incrementally, and then provide a tiny perturbation to the column. This can be done by hitting the table on which the apparatus rests. The smallest load at which the column buckles will be your experimentally determined buckling load. You must repeat the same experiment multiple times to measure the uncertainty involved, and to ensure repeatability.

Once you have obtained several such readings for different boundary conditions, compare them to the theoretical formula stated in the “Background section”. Seek help from the lab staff and TAs if you need to.

Task 2:

Write a report on the entire exercise. The report must not exceed 3 pages. You can tailor your report as per the following proposed structure, but you are free to deviate from it if you wish so.

- 1.) **INTRODUCTION:** State your objectives, and answers to questions 1-5 stated in the “Background” section above.
- 2.) **PROCEDURE:** Describe the details of the procedure used to conduct the experiment.
- 3.) **OBSERVATIONS:** Report all the experimental data recorded. Describe your observations, insights, any calculations you perform. Use plots and tables to compare

experiments with theory. Does your experimental data agree with the theoretical prediction? If not, explain why.

4.) **CONCLUSIONS:** Describe what you understood from this entire exercise.